



Semester 1 Examinations 2017

Exam Code(s) 1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam(s) Higher Diploma in Software Design and Development
Higher Diploma in Software Design and Development,
Industry Stream
Masters in Information Technology
Masters in Software Design and Development
Masters in Digital Media
Masters Clinical Research

Module Code(s) CT511
Module(s) Database Development

Paper No.
Repeat Paper

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Instructions: Answer **questions 1 and 2** and 2 other questions
All questions carry equal marks

Duration 2 Hours
No. of Pages 7
Discipline(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper

Other Materials

Release to Library: Yes ☒ No ☐

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Question 1

- a) Create an **EERD** for the following:

NUI Ragoon is a new institute of higher education. It has many colleges, such as Arts, Engineering, Medicine and Science. Because it's a new institute, it plans for future expansion. Each college has a college ID, head and phone number. College lecturers are assigned to only one college.

The Arts college has a dedicated landline.

The Engineering college engages with a number of external auditors as some of its programs require this. This college needs their pps, name and contact details. Meanwhile the Science college has its own specific opening hours.

Colleges have three types of lecturer only; contract, junior and senior lecturer. The contract lecturer has a contract start and end date.

All lecturers regardless of their contract type is identified by ID, name, address and date of birth. In addition to lecturers, colleges have different categories of admin; admin grade 1, admin grade 2 and admin grade 3. There is a future plan to allow for grades 4 and 5.

Some lecturers teach courses and engage in research projects, while other lectures just engage in full time research. Research projects generally tend to have at least one lecturer assigned to it.

All research projects must include the following; project Id, description, date started, estimated cost, project code and total cost to date.

(17 marks)

- b) Indicate the normal form and decompose to 3NF if not in 3NF

CLASS (COURSE-No, SECTION-No)

CLASS (COURSE-No, SECTION-No, ROOM)

CLASS (COURSE-No, SECTION-No, ROOM, CAPACITY)

(6 marks)

- c) What is meant by the term meta-data (use examples)?

(2 marks)

Question 2

- (a) Write the SQL for a MYSQL Server DB to :

1. Create a table called **courses** with the following fields; coursecode can contain text of 5 characters, description can contain text of 25 characters, coordinator can contain text of 25 characters while department can contain text of 10 characters. Coursecode is the primary key.
2. Ensure that module (which is 5 characters in length) in the **students table** (see next page) can accept null values and is also the foreign key which links the **students table** with the **courses table** via coursecode.

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students table

SID	name	module	mark	discipline
S101	A Lynch	CT101	45.00	English
S102	M Duignan	CT101	75.00	Philosophy
S103	M Peters	CT103	55.00	Philosophy
S104	A Lynch	NULL	22.00	Philosophy

3. Insert the following data into the **courses table**;

courses table

coursecode	description	coordinator	department
CT101	Programming	J Hill	IT
CT102	Programming	J Hill	IT
CT103	Algebra	C Byrne	Maths
CT104	Algebra	C Fox	Maths

4. Find the total number of students in each department and display the following:

The total number of students in each department
number of students in the IT department is 2
number of students in the Maths department is 2

5. Check whether or not the student's mark is less than the average of all marks. The following should be displayed:

module	mark	Mark Commentary
CT101	45.00	The mark is less than the average mark of 48.50
CT101	55.00	Well done
CT103	72.00	Well done
NULL	22.00	The mark is less than the average mark of 48.50

6. Find the unique disciplines where the **SID** is equal to any of the SIDs returned where the mark is less than or equal to 60. Display the following:

Unique disciplines
English
Philosophy

7. Display the following for all records whose coordinator name does not begin with a J:

coursecode	description	coordinator	department
CT103	Algebra	C Byrne	Maths
CT104	Algebra	C Fox	Maths

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8. Display all the students names, modules, descriptions and departments regardless of whether or not they are associated with a module (use table aliases).

name	module	description	department
A Lynch	CT101	Programming	IT
M Duignan	CT101	Programming	IT
M Peters	CT103	Algebra	Maths
A Lynch	NULL	NULL	NULL

9. Display only those records where the mark is between the average and maximum mark.

SID	name	module	mark	discipline
S102	M Duignan	CT101	55.00	Philosophy
S103	M Peters	CT103	72.00	Philosophy

10. Display the name and mark of the **top** record where the mark is greater than or equal to **ALL** of the minimum student marks grouped by discipline and presented in **descending** order.

name	mark
M Peters	72.00

11. Use self join, to display the following from the **students** table where the module is not the same and the marks of that record are greater than that of the one being compared.

name	discipline	module	mark	name	discipline	module	mark
M Peters	Philosophy	CT103	72.00	A Lynch	English	CT101	45.00
M Peters	Philosophy	CT103	72.00	M Duignan	Philosophy	CT101	55.00

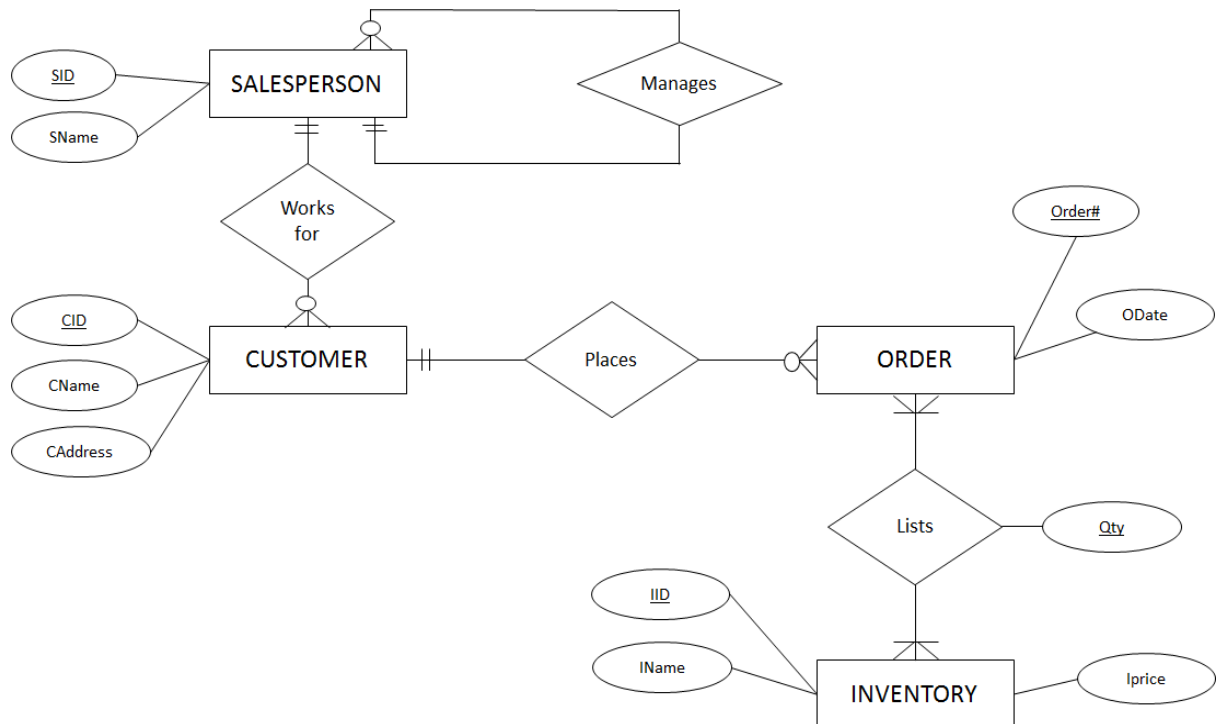
12. Create a view called **studentmodule** which contains the student name, module name and course coordinator of all courses which contain students.

(25 marks)

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Question 3

(a) Map the following to relations using the short hand approach.



(10 marks)

(b) What is a **view** (use examples)? Outline advantages and disadvantages of views.

(8 marks)

(c) Compare and contrast **left** and **right joins**. Use sample tables and sample SQL code to highlight the difference.

(7 marks)

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Question 4

- a) Bring the following table to 2NF explaining why it is not in 2NF (use the 5 step approach).

CourseID	CourseName	ProfID	StudentID	SName	StartDate
C101	Programming	P1	S123	M Fox	23/5/2017
C101	Programming	P1	S124	P Liston	12/5/2017
C102	Algebra	P1	S123	M Fox	12/9/2017
C103	Maths	P3	S125	A Hill	18/12/2017
C103	Maths	P2	S123	M Fox	13/5/2018

(15 marks)

- b) Outline 4 issues with computer based systems which led to DBMSs (use examples).

(6 marks)

- c) Explain what is meant by the terms '**Data Dependence**' and '**Data Atomicity**' (use examples).

(4 marks)

Question 5

- a) Bring the following table to 3NF explaining why it is not in 3NF. Also explain why it is in 2NF.

BookID	GenreID	GenreType	Price
1	1	Gardening	12.00
2	2	Sports	15.99
3	1	Gardening	9.99
4	3	Travel	12.88
5	2	Sports	16.43

(10 Marks)

- b) Compare and contrast the 4 **data integrity constraints** (use examples).

(8 Marks)

- a) Draw and then discuss the '**Three Schema Architecture**'.

(7 Marks)

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Question 6

- a) Compare and contrast the 3 options for **denormalisation** (use examples). (10 Marks)
- b) Discuss 4 ways of controlling unauthorized access to the DBMS (use examples). (8 Marks)
- c) What is a stored procedure in MySQL? Write a stored procedure called **westerncustomers** which prints out the first and last names and balance from the **customers** table where the county is either Galway or Mayo or Sligo.

Customers table

FirstName	LastName	County	Balance
Ann	Moody	Mayo	1245
Michael	Davitt	Tipperary	344
John	Mullholland	Sligo	454
Mary	Tippins	Galway	234234
Lili	Duggan	Longford	3423423
Suzy	McDermott	Galway	472384

(7 Marks)